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Cambridge IGCSE Combined and Co-ordinated Sciences

B1 Cells and Organisms

Condensed Version — Key Concepts and High-Probability Exam Questions

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Cambridge IGCSE Biology · The most important concepts and exam questions for your test

Must-Know Concepts

The Seven Characteristics of Living Organisms

- **MRS GREN**: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition. **All** organisms show **all seven**.
- **Not** shown by all organisms: **photosynthesis**, **breathing**, a **blood system** and a **nervous system**.
- **Respiration is not breathing:** respiration is chemical reactions in **every living cell**, day and night, releasing energy from glucose. Plants and bacteria respire but do not breathe.
- **Growth must be permanent:** drinking water makes you heavier for a while, but that is not growth. **Dry mass** is more reliable than fresh mass because the amount of water in an organism keeps changing.
- **Metabolism** makes waste products, some of them toxic, so every organism must **excrete**, e.g. breathing out carbon dioxide made by respiration, or getting rid of urea in urine.

Cell Structures and Their Functions

- **All organisms are made of cells.** Cells are very small, so large organisms contain millions of cells. **New cells** are only made by the **division of existing cells**.

Structure	Function	Animal	Plant	Bacterium
Cell membrane	Partially permeable; controls what enters and leaves the cell	Yes	Yes	Yes
Cell wall	Fully permeable; protects and supports the cell; stops it bursting when it takes up water	No	Yes	Yes
Cytoplasm	Jelly where many metabolic reactions (the chemical reactions of life) take place	Yes	Yes	Yes
Nucleus	Contains DNA in chromosomes; the DNA determines which proteins the cell makes	Yes	Yes	No
Ribosomes	Make proteins (protein synthesis)	Yes	Yes	Yes
Mitochondria	Aerobic respiration, releasing energy from glucose	Yes	Yes	No
Chloroplasts	Chlorophyll absorbs energy from sunlight for photosynthesis	No	Often	No
Large permanent vacuole	Stores cell sap; helps the cell keep its shape	No	Yes	No
Plasmids	Small extra circles of DNA	No	No	Often

- **Key idea:** plant cell = animal cell structures + **cellulose** cell wall + large permanent vacuole + (often) chloroplasts, which often contain **starch grains**. Animal cells never contain starch grains and have only small vacuoles called **vesicles**.
- **The cell wall is the reliable sign of a plant cell:** many plant cells, such as root hair cells, have no chloroplasts.

- **Membrane and wall:** the cell membrane is a very thin layer of protein and fat. In a plant cell it is pressed tightly against the inside of the cell wall, so it is hard to see. The wall is fully permeable; only the partially permeable membrane controls what enters and leaves.
- **Nucleus:** the genetic information is stored on **chromosomes**, which are made of **DNA** and inherited from the organism's parents. **DNA decides, ribosomes build:** ribosomes use the DNA's instructions to link amino acids together in a particular sequence to make a protein.
- **Mitochondria release energy** from glucose by aerobic respiration (glucose + oxygen → carbon dioxide + water). Cells that need more energy, such as muscle cells, have more mitochondria.
- **A cell that makes a lot of protein** needs many ribosomes to make the protein and many mitochondria to release the energy needed.
- **Microscopes:** a very good light microscope magnifies about ×1500; you need an electron microscope to see **ribosomes**, and to see **mitochondria** clearly.
- **Spotting structures:** the cell wall is a thick outer layer, often a double line; the nucleus is usually the darkest part of a stained cell; the large vacuole is a big, clear space; mitochondria are small sausage shapes with stripes; ribosomes are tiny dots seen only in electron micrographs.

Bacterial Cells

- Bacteria are **unicellular** and have **prokaryotic** cells, with **no nucleus** ('pro' means before and 'karyotic' means nucleus). Prokaryotic cells appeared on Earth millions of years before cells with nuclei.
- Instead of chromosomes in a nucleus, a bacterium has a **circle of DNA** in its cytoplasm, sometimes called a bacterial chromosome. It does the same job as in other cells: it provides the instructions for making proteins.
- Bacteria often have one or more **plasmids**: smaller circles of DNA. Scientists use plasmids in the **genetic modification** of cells and organisms.
- The **cell wall** is made of **peptidoglycan, not cellulose**, but does the same job as in plants: it supports the cell and stops it bursting. A partially permeable **cell membrane** is pressed tightly against the inside of the wall and controls what enters and leaves.
- **Cytoplasm and ribosomes** do the same jobs as in animal and plant cells. Bacteria have **no mitochondria and no chloroplasts**.

Specialised Cells

- A **specialised cell** has a structure modified to help it carry out a particular function effectively.

Cell (where found)	Function	How its structure helps
Root hair cell (near root tips)	Absorbs water and mineral ions	Long, thin root hair gives a large surface area
Palisade mesophyll cell (leaves)	Photosynthesis	Packed with chloroplasts to absorb a lot of light energy
Red blood cell (blood of mammals)	Transports oxygen	Full of haemoglobin, which carries oxygen; no nucleus, leaving more room for haemoglobin
Ciliated cell (lining trachea and bronchi)	Cilia move mucus upwards	Cilia beat to sweep mucus, with trapped dust and bacteria, upwards
Neurone (nervous system)	Conducts electrical impulses	Very long, thin fibre carries impulses over long distances
Sperm cell (made in testes)	Male gamete	Tail for swimming to the egg; many mitochondria release energy for swimming

Cell (where found)	Function	How its structure helps
Egg cell (made in ovaries)	Female gamete	Large, with a store of food for the embryo

Levels of Organisation

- **cells** → **tissues** → **organs** → **organ systems** → **organism** (a complete living thing), for example: stomach lining cell → stomach lining tissue → stomach → digestive system → human.
- A **tissue** is made of **similar** cells; an **organ** is made of **different** tissues.

Key Terms

Term	Definition
Movement	An action by an organism or part of an organism causing a change of position or place
Respiration	The chemical reactions in cells that break down glucose and other nutrient molecules to release energy for the cell to use
Sensitivity	The ability to detect changes in the internal or external environment and respond to them
Growth	A **permanent** increase in size and dry mass, by the growth of cells and the addition of new cells
Reproduction	The processes that make new organisms of the same species
Excretion	The removal from the body of the waste products of metabolism (unwanted or toxic substances) and of substances in excess of requirements
Nutrition	Taking in substances from the environment and using them for energy, growth and development
Metabolism	All the chemical reactions that take place in an organism's cells, including respiration
Cell	The smallest unit from which all organisms are made
Unicellular	Made of just a single cell; bacteria and yeast are unicellular
Multicellular	Made of many cells; humans and plants are multicellular
Partially permeable	Allows some molecules and ions to pass through, but not others
Fully permeable	Allows all molecules and ions to pass through it
Aerobic respiration	Chemical reactions in mitochondria that use oxygen to break down glucose and other nutrient molecules to release energy for the cell to use
Prokaryotic cells	Cells with no nucleus; bacteria have prokaryotic cells
Plasmids	Small, circular molecules of DNA found in many prokaryotic cells in addition to the main, much larger circle of DNA
Tissue	A group of **similar** cells that work together to perform a particular function
Organ	A structure made up of a group of tissues, working together to perform specific functions

Term	Definition
Organ system	A group of organs with related functions, working together to perform body functions
Magnification	How many times larger an image is than the actual object

Magnification: Equations, Units and Method

- **magnification = size of image ÷ size of actual object**
- **actual size = size of image ÷ magnification**
- **size of image = actual size × magnification**
- Put both sizes in the **same units** before you divide. Write a magnification with a **× sign** and **no units**: ×500, never 500 mm.
- **Show every step** (each can earn a mark): measure the image in mm and write it down, convert to the same units, write the equation, substitute, then give the answer and check that it is sensible.
- **Units:** 1 cm = 10 mm; **1 mm = 1000 µm**; 1 µm = 1×10^{-6} m; 1 m = 10^{6} µm. Smaller unit, bigger number (multiply): 8 cm = 80 mm, 2.5 mm = 2500 µm, 0.002 m = 2000 µm. Bigger unit, smaller number (divide): 40 µm = 0.04 mm.
- **Worked example:** a cheek cell is 30 mm wide in a photomicrograph and 60 µm wide in real life. Convert first: 30 mm × 1000 = 30 000 µm. Then magnification = 30 000 µm ÷ 60 µm = **×500**.

High-Probability Exam Questions

Question 1

Identify the characteristic which is **not** shown by all living organisms: A excretion, B movement, C photosynthesis, D respiration. (1 mark)

Answer:

- **C, photosynthesis.** Only plants (and some other organisms such as algae) photosynthesise; animals, fungi and most bacteria do not. Excretion, movement and respiration are shown by all living organisms.

Question 2

The tongue-eating louse, a parasite of large fish, detects chemicals on the gills of a fish and crawls from the gills into the fish's mouth. It sucks blood from the tongue until the tongue falls off, then takes its place, feeding on the fish's food.

a) Define **nutrition**. [2]

b) Apart from nutrition, explain how **two** features of the louse are characteristics of all living organisms. [2]

c i) Define **reproduction**. [2]

c ii) Suggest **one** reason why scientists know very little about reproduction in this species. [1]

(7 marks)

Answer:

- a) Taking in substances (materials) from the environment
- a) and using them for energy, growth and development
- b) Sensitivity: it detects chemicals on the fish's gills, so it detects and responds to a change in its environment
- b) Movement: it crawls from the gills into the fish's mouth, so it changes its position
- c i) The processes that make new organisms

- c i) of the same species (the same kind of organism)
- c ii) Any one sensible reason, e.g. it lives hidden inside the mouths of fish in the sea, so it is very hard to observe

Question 3

A student claimed that plants show fewer of the characteristics of living things than animals. Explain why this claim is wrong. (8 marks)

Answer:

- Plants show all seven characteristics; some just happen more slowly or less obviously than in animals
- Movement: plants slowly move parts of themselves, e.g. many flowers open by day and close at night
- Respiration: every living plant cell respires, day and night, releasing energy from glucose
- Sensitivity: plants detect and respond to changes in their environment, e.g. shoots grow towards light
- Growth: plants grow throughout their lives, with a permanent increase in size and dry mass
- Reproduction: plants make new plants of the same species, e.g. from seeds
- Excretion: plants get rid of waste products of metabolism, such as oxygen from photosynthesis and carbon dioxide from respiration
- Nutrition: plants make their own food by photosynthesis and take in water, carbon dioxide and mineral ions

Question 4

Name the part of a cell which has each of these functions: a) makes proteins; b) contains the information for which proteins to make; c) photosynthesis; d) prevents the cell bursting when it takes up water; e) stores a solution of sugars and other solutes; f) controls what enters and leaves the cell. (6 marks)

Answer:

- a) Ribosome
- b) Nucleus (the DNA in its chromosomes carries the information)
- c) Chloroplast
- d) Cell wall
- e) Vacuole (the large permanent vacuole of a plant cell, which contains cell sap)
- f) Cell membrane

Question 5

Which of the structures that you named in Question 4 are found in both animal and plant cells? (3 marks)

Answer:

- Ribosome
- Nucleus
- Cell membrane

The chloroplast and the cell wall are found only in plant cells. Animal cells have only small vacuoles called vesicles, not a large permanent vacuole.

Question 6

A diagram of a cell from a plant leaf has four labels. A: the outer double line that forms the boundary of the cell. B: a small oval structure at the left-hand side of the cell. C: the narrow layer between the outer

boundary and the large central space. D: the large central space that fills most of the cell. Identify which labelled part contains DNA. (1 mark)

Answer:

- **B**, the nucleus, which contains DNA in the form of chromosomes (A is the cell wall, C is the cytoplasm and D is the large permanent vacuole)

Question 7

A jellyfish belongs to the animal kingdom. In a photograph, a jellyfish is 89 mm across its widest part. The actual jellyfish is 50 mm in diameter.

a) Calculate the magnification of the photograph. [3]

b) State **three** parts that you would find in a cell in a jellyfish **and** outline the function of each part. [6]

(9 marks)

Answer:

- a) magnification = size of image ÷ size of actual object
- a) = 89 mm ÷ 50 mm
- a) = 1.78, so the magnification is **×1.8** to 2 significant figures (× sign, no units)
- b) Part: nucleus
- b) Function: contains DNA, which carries the information about which proteins the cell makes
- b) Part: cell membrane
- b) Function: partially permeable; controls what enters and leaves the cell
- b) Part: mitochondrion
- b) Function: the site of aerobic respiration, which releases energy from glucose

Cytoplasm (where many metabolic reactions take place) and ribosomes (where proteins are made) are also correct parts. A jellyfish is an animal, so its cells have no cell wall, no chloroplasts and no large permanent vacuole.

Question 8

A diagram of a flower is 8 cm across. The magnification of the diagram is ×4. Identify the actual width of the flower: A 3.2 mm, B 20 mm, C 0.5 cm, D 32 cm. (1 mark)

Answer:

- **B, 20 mm.** Actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm, and 2 cm × 10 = 20 mm. D (32 cm) comes from multiplying instead of dividing.

Question 9

An electron micrograph of a mitochondrion is magnified 20 000 times; the maximum length of the mitochondrion on the micrograph is 83 mm. a) How many micrometres are there in 1 mm? b) Convert 83 mm to µm. c) Calculate the real size of the mitochondrion in µm. d) How many of these mitochondria could you line up end to end between two of the mm marks on a ruler? (4 marks)

Answer:

- a) 1000 µm (to convert mm to µm, multiply by 1000)
- b) 83 mm × 1000 = 83 000 µm
- c) Real size = size of image ÷ magnification = 83 000 µm ÷ 20 000 = **4.15 µm**
- d) Two mm marks are 1 mm = 1000 µm apart, so 1000 µm ÷ 4.15 µm = 240.96: about **241** mitochondria (240 whole ones would fit)

Common Mistakes to Avoid

- **Writing that mitochondria make energy:** they **release** energy from glucose by aerobic respiration.

- **Confusing respiration with breathing:** every living cell respires, but plants and bacteria do not breathe.
- **Leaving "permanent" out of the definition of growth.**
- **Saying the cell wall controls what enters the cell:** it is fully permeable; the partially permeable membrane does this.
- **Calling a cell without chloroplasts an animal cell** (look for the cell wall), or giving animal cells a large vacuole (they have only small vesicles).
- **Naming DNA when the question asks for a part of the cell:** the part is the **nucleus**, which contains the DNA.
- **Magnification slips:** multiplying instead of dividing to find the actual size; mixing mm and µm in one calculation (the answer is 1000 times too big or too small); putting units on a magnification.

Last-Minute Checklist

- I can define each of the seven characteristics (MRS GREN) in exam wording.
- I can explain why respiration is not breathing and why growth must be permanent.
- I can name the structures in animal, plant and bacterial cells and state their functions.
- I can describe a bacterial cell: no nucleus, a circle of DNA, plasmids, and a cell wall that is not made of cellulose.
- I can explain partially permeable versus fully permeable.
- I can identify cell structures in diagrams and micrographs.
- I can explain why a cell that makes a lot of protein needs many ribosomes and mitochondria.
- I can state the function of each specialised cell and how its structure helps.
- I can order cell, tissue, organ, organ system and organism, with an example of each.
- I can convert between cm, mm and µm and calculate magnification, actual size and image size, showing my working.